11.1 FUTURE SCOPE :

The HematoVision project has strong potential for future enhancements and expansion in the field of AI-based medical diagnostics. Although the current system focuses on classifying four types of blood cells using transfer learning, several improvements can be implemented to increase its accuracy, scalability, and real-world impact.

In the future, the model can be trained with a larger and more diverse dataset collected from multiple laboratories. This will improve generalization across different imaging conditions, microscopes, and staining techniques. Expanding the dataset will also help reduce bias and enhance prediction reliability.

The system can be extended to detect additional blood cell abnormalities and diseases such as advanced leukemia classification, malaria parasite detection, and other hematological disorders. Instead of only classification, the model can be enhanced to perform object detection and segmentation to identify and highlight abnormal cells within an image.

Explainable AI (XAI) techniques can be integrated to improve transparency and trust. Visual explanation tools such as heatmaps can show which regions of the image influenced the model's decision, increasing confidence among healthcare professionals.

Deployment can be expanded to cloud-based platforms for wider accessibility. A mobile application version can also be developed to enable remote diagnostics in rural or low-resource healthcare settings. Integration with hospital management systems and laboratory information systems can further automate workflow processes.

Additionally, incorporating real-time data analytics and continuous learning mechanisms will allow the model to update itself with new data and maintain high performance over time.

In conclusion, the future scope of HematoVision lies in expanding its capabilities, improving explainability, increasing accessibility, and integrating advanced AI techniques to transform it into a comprehensive intelligent diagnostic support system for modern healthcare.